

5.4 Security Threats and Mitigation Techniques

سیکیورٹی خطرہ وہ عمل یا کمزوری ہے جو نظام یا ڈیٹا کو نقصان پہنچاے۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 10 of 15

★ Golden

5.4 Security Threats and Mitigation Techniques ★ Golden

LEARN IT

A **security threat** is any action, event or weakness that can **harm a computer system or its data**. **Mitigation techniques** are protective measures used to prevent threats or reduce their impact. This section names the common threats; later sections show how to spot them and how to reduce the damage.

#	Threat	What it is	Example
1	Malware	Harmful software designed to damage a computer, steal data, or gain unauthorised access. Includes viruses, worms, spyware, ransomware and Trojans .	Installing cracked software that secretly corrupts files, or a "free game" USB that encrypts a college lab's coursework (ransomware).
2	Phishing	A social-engineering trick that deceives users into sharing personal information (passwords, bank details, OTPs) using fake emails or websites that look genuine.	An SMS that looks like JazzCash or HBL asking you to "verify" by typing your OTP on a fake page.

3	Unauthorised access	Entering a system, account or file without permission. Often caused by weak passwords, shared passwords, or leaving devices unlocked.	A classmate using an unlocked lab PC to open someone else's Google Classroom.
4	Data theft / identity theft	Data theft is stealing or copying data without permission. Identity theft is using another person's stolen details (name, CNIC, bank details) to pretend to be them online and commit fraud.	A leaked list of CNICs used to open a SIM or a wallet in someone else's name.
5	Denial of Service (DoS)	A website or system is flooded with thousands of fake requests, causing it to slow down or crash, so real users cannot access the service.	Hackers flooding an online exam portal so students cannot log in on BISE result day or during an e-test.
6	Human error	Mistakes by legitimate users: accidentally deleting files, sending confidential data to the wrong email, or falling for a simple scam.	A teacher emailing the whole marksheet to the wrong parent; a student forwarding an OTP "to help the bank."

Malware types in one line each: a **virus** attaches to files and spreads when you share them; a **worm** spreads by itself across a network; **spyware** secretly watches you; **ransomware** locks your files and demands money; a **Trojan** pretends to be useful software but hides harm.

Social engineering means tricking a **person**, not breaking a lock with maths. Phishing is the most common exam example.

Phishing

Fake bank email / SMS
steals password or OTP

DoS attack

Flood of fake requests
server overload / crash

Identity theft

Stolen CNIC / details
fraud in your name

Three high-yield threats: phishing (fake messages), DoS (server overload), identity theft (stolen CNIC or bank details).

WORKED EXAMPLE

Worked example: classify the incident.

1. Bilal downloads a "cracked" Photoshop. Next day his project files are unreadable and a note asks for Bitcoin. → **Malware** (ransomware), started by cracked software.
2. Ayesha gets an email "Your Meezan account will close in 2 hours. Click here and send OTP." The link is meezn-pk.tk, not the real bank. → **Phishing**.
3. On e-test morning the college portal spins forever; the admin graph shows 200,000 requests from one pattern of fake traffic. → **DoS**.
4. Someone opens a JazzCash wallet using Ahmed's CNIC copy from a leaked form. → **Identity theft** (after data theft).
5. The lab assistant emails the password list to the whole staff by choosing the wrong group. → **Human error**.

REMEMBER IT

Malware = bad program. Phishing = fake message. DoS = flood. Identity theft = stolen 'you'. Human error = honest mistake. Cracked software and unknown USBs are the usual door for malware in Pakistani colleges.

COMMON MISTAKE

Mistake: Calling every cyber problem a "virus", or mixing phishing with malware.

Fix: A virus is only one kind of malware. Phishing is a **trick against the user**; it may not install any file at all. DoS does not steal your password — it makes the service unusable for everyone.

سیکیورٹی خطرہ وہ عمل یا کمزوری ہے جو نظام یا ڈیٹا کو نقصان پہنچاے۔ میلویئر نقصان دہ سافٹ ویئر ہے (وائرس، ورم، سپائی ویئر، رینسم ویئر، ٹروجن)۔ فشننگ جعلی ای میل یا ویب سائٹ سے پاس ورڈ اور او ٹی پی چھینتی ہے۔ ڈو ایس (DoS) جعلی درخواستوں کی بارش سے سائٹ کریش کر دیتا ہے۔ دیگر خطرات: غیر مجاز رسائی، ڈیٹا/شناخت کی چوری، اور انسانی غلطی۔

EXAM TIP

Expect "Define a security threat", "Explain malware / phishing / DoS with examples", and "Differentiate data theft and identity theft." List all six threats if the question says "common security threats." For DoS, write: flood of fake requests → slowdown or crash → real users blocked, and quote the online exam portal. For phishing, write social engineering + fake genuine-looking email/site + passwords/OTPs.

CHECK YOURSELF

1. Name five types of malware listed in the source.
2. What does a DoS attack do to a website?
3. How is identity theft different from data theft?

Answers: 1. Viruses, worms, spyware, ransomware, Trojans. 2. Floods it with thousands of fake requests so it slows down or crashes and real users cannot access it. 3. Data theft is stealing/copying data; identity theft is using stolen personal details (name, CNIC, bank) to pretend to be that person and commit fraud.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Malware is:

- a. Malicious software (virus, worm, trojan, ransomware...)
- b. A friendly backup
- c. A primary key
- d. A caption

MCQ 2. Phishing is:

- a. Tricking you into giving secrets (fake mail/site)
- b. A fishing sport in the syllabus
- c. A sort
- d. A chart

MCQ 3. DoS aims to:

- a. Make a service unavailable by flooding it
- b. Encrypt with your key
- c. Fix UX
- d. Normalise tables

MCQ 4. A threat is:

- a. Anything that can harm system or data
- b. Only earthquakes
- c. Only NAND delay
- d. Only slow Wi-Fi in exams

MCQ 5. This is ★: name, one example, one mitigation.

- a. The 3-mark recipe
- b. Only the name
- c. Only Python
- d. Only Figma

MCQ 6. A fake BIEK result link is likely:

- a. Phishing
- b. A K-map
- c. A stack
- d. A median

Extra short questions

1. Define malware, phishing, DoS.
2. One extra threat from the notes (theft/identity).
3. A student-level mitigation for phishing.
4. Why USB etiquette?

Extra long question

1. Table of four threats: what it is, a Sindh-college example, a sign, a mitigation. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Malware: bad software. Phishing: social trick. DoS: flood to outage.
2. Data/identity theft — using someone's CNIC/account.
3. Check URL, don't tap unknown SMS, official portal only.
4. Worms travel on sticks.

Long

1. Malware, phishing, DoS, theft. Signs: slowness, fake from-addresses, site down, unknown logins. Mitigations: updates/AV, URL check, backups+capacity, MFA.

Next lecture: 5.4.2 Identify Security Threats